Reverse-check review package — `HeavyN_gen`

An independent, **blank-slate** agent instance reconstructed the physics below from the sanitized `.fr` alone (no paper, no metadata, no conversation history). A second fresh instance then compared the reconstruction against the source paper. The final verdict belongs to the human reviewer — sign off at the bottom.

item	value
model file	`HeavyN/model/HeavyN_gen.fr`
original model name	`HeavyN_gen` (hidden from the agent)
paper	HeavyN/text/1602.06957.txt
blindness scope	blank slate w.r.t. paper, authors, model name, comments and prose labels — NOT w.r.t. field/parameter symbol names, which are kept so the reviewer can map the reconstruction back

Verbatim Lagrangian terms (from the `.fr`)

These are the terms the reconstruction must account for, quoted unmodified. Check each against its reconstructed LaTeX form below.

`LNKin` (:=)

```
I/2 N1bar[s1].Ga[v,s1,s2].del[N1[s2],v] - 1/2 mN1 N1bar[s1] N1[s1] + I/2
 ↪  N2bar[s1].Ga[v,s1,s2].del[N2[s2],v] - 1/2 mN2 N2bar[s1] N2[s1] + I/2
 ↪  N3bar[s1].Ga[v,s1,s2].del[N3[s2],v] - 1/2 mN3 N3bar[s1] N3[s1]
```

`LNCCbare` (:=)

```
gN/Sqrt[2]*(VeN1*N1bar.W[m].ProjM[m].e + VmuN1*N1bar.W[m].ProjM[m].mu +
 ↪  VtaN1*N1bar.W[m].ProjM[m].ta) + gN/Sqrt[2]*(VeN2*N2bar.W[m].ProjM[m].e +
 ↪  VmuN2*N2bar.W[m].ProjM[m].mu + VtaN2*N2bar.W[m].ProjM[m].ta) +
 ↪  gN/Sqrt[2]*(VeN3*N3bar.W[m].ProjM[m].e + VmuN3*N3bar.W[m].ProjM[m].mu +
 ↪  VtaN3*N3bar.W[m].ProjM[m].ta)
```

`LNCC` (:=)

```
LNCCbare + HC[LNCCbare]
```

`LNNCBare` (:=)

```
1/2*gN/cw*(VeN1*N1bar.Z[m].ProjM[m].ve + VmuN1*N1bar.Z[m].ProjM[m].vm +
 ↪  VtaN1*N1bar.Z[m].ProjM[m].vt) + 1/2*gN/cw*(VeN2*N2bar.Z[m].ProjM[m].ve +
 ↪  VmuN2*N2bar.Z[m].ProjM[m].vm + VtaN2*N2bar.Z[m].ProjM[m].vt) +
 ↪  1/2*gN/cw*(VeN3*N3bar.Z[m].ProjM[m].ve + VmuN3*N3bar.Z[m].ProjM[m].vm +
 ↪  VtaN3*N3bar.Z[m].ProjM[m].vt)
```

`LNNC` (:=)

```
LNNCBare + HC[LNNCBare]
```

`LNHbare` (:=)

```
-gN*mN1/(2*MW)*(VeN1*N1bar.ProjM.ve H + VmuN1*N1bar.ProjM.vm H + VtaN1*N1bar.ProjM.vt H) -
 ↪  gN*mN2/(2*MW)*(VeN2*N2bar.ProjM.ve H + VmuN2*N2bar.ProjM.vm H + VtaN2*N2bar.ProjM.vt H)
 ↪  - gN*mN3/(2*MW)*(VeN3*N3bar.ProjM.ve H + VmuN3*N3bar.ProjM.vm H + VtaN3*N3bar.ProjM.vt H)
```

LNH (:=)

```
LNHbare + HC[LNHbare]
```

LNGbare (:=)

```
I*gN*mN1/(2*MW)*(VeN1*vebar.ProjP.N1 G0 + VmuN1*vmbar.ProjP.N1 G0 + VtaN1*vtbar.ProjP.N1 G0)
 ↪   + I*gN*mN2/(2*MW)*(VeN2*vebar.ProjP.N2 G0 + VmuN2*vmbar.ProjP.N2 G0 +
 ↪   VtaN2*vtbar.ProjP.N2 G0) + I*gN*mN3/(2*MW)*(VeN3*vebar.ProjP.N3 G0 + VmuN3*vmbar.ProjP.N3
 ↪   G0 + VtaN3*vtbar.ProjP.N3 G0) + I*gN*mN1/(Sqrt[2]*MW)*(VeN1*ebar.ProjP.N1 GPbar +
 ↪   VmuN1*mubar.ProjP.N1 GPbar + VtaN1*tabar.ProjP.N1 GPbar) +
 ↪   I*gN*mN2/(Sqrt[2]*MW)*(VeN2*ebar.ProjP.N2 GPbar + VmuN2*mubar.ProjP.N2 GPbar +
 ↪   VtaN2*tabar.ProjP.N2 GPbar) + I*gN*mN3/(Sqrt[2]*MW)*(VeN3*ebar.ProjP.N3 GPbar +
 ↪   VmuN3*mubar.ProjP.N3 GPbar + VtaN3*tabar.ProjP.N3 GPbar)
```

LNG (:=)

```
LNGbare + HC[LNGbare]
```

LN (:=)

```
LNKin + LNCC + LNNC + LNH + LNG
```

LFull (:=)

```
LSM + LN
```

Blank-slate reconstruction

Reconstructed Physics from `sanitized.fr`

Lagrangian

The file defines

$$\texttt{LFull} = \mathcal{L}_{\rm SM} + \texttt{LN},$$

with

$$\texttt{LN} = \texttt{LNKin} + \texttt{LNCC} + \texttt{LNNC} + \texttt{LNH} + \texttt{LNG}.$$

The new fermions N_i, $i = 1, 2, 3$, are self-conjugate neutral spin-$\frac{1}{2}$ fields. Since they are declared with $Q = 0$, $Y = 0$, no color index, and no weak-isospin index, their covariant derivative contains no gauge connection:

$$D_\mu N_i = \partial_\mu N_i.$$

Let $\ell_\alpha = (e, \mu, \tau)$, $\nu_\alpha = (\nu_e, \nu_\mu, \nu_\tau)$, and $V_{\alpha i} \in \mathbb{R}$ denote the parameters

$$V_{\alpha i} = \begin{pmatrix} \texttt{VeN1} & \texttt{VeN2} & \texttt{VeN3} \\ \texttt{VmuN1} & \texttt{VmuN2} & \texttt{VmuN3} \\ \texttt{VtaN1} & \texttt{VtaN2} & \texttt{VtaN3} \end{pmatrix}.$$

The internal coupling is

$$g_N = \frac{e}{s_w}.$$

Here $P_L = \frac{1-\gamma^5}{2}$, $P_R = \frac{1+\gamma^5}{2}$. FeynRules `ProjM` is expanded as P_L, and `ProjP` as P_R.

LNKin

$$\boxed{\texttt{LNKin} = \sum_{i=1}^{3} \left[\frac{i}{2}\, \overline{N}_i \gamma^\mu \partial_\mu N_i - \frac{1}{2} m_{N_i}\, \overline{N}_i N_i \right]}$$

LNCC

The bare term in the file is

$$\texttt{LNCCbare} = \frac{g_N}{\sqrt{2}} \sum_{i=1}^{3} \sum_{\alpha=e,\mu,\tau} V_{\alpha i}\, \overline{N}_i \gamma^\mu P_L \ell_\alpha\, W_\mu .$$

The Lagrangian term is

$$\boxed{\texttt{LNCC} = \frac{g_N}{\sqrt{2}} \sum_{i=1}^{3} \sum_{\alpha=e,\mu,\tau} V_{\alpha i}\, \overline{N}_i \gamma^\mu P_L \ell_\alpha\, W_\mu + \text{h.c.}}$$

LNNC

The bare term in the file is

$$\texttt{LNNCBare} = \frac{g_N}{2c_w} \sum_{i=1}^{3} \sum_{\alpha=e,\mu,\tau} V_{\alpha i}\, \overline{N}_i \gamma^\mu P_L \nu_\alpha\, Z_\mu .$$

The Lagrangian term is

$$\boxed{\texttt{LNNC} = \frac{g_N}{2c_w} \sum_{i=1}^{3} \sum_{\alpha=e,\mu,\tau} V_{\alpha i}\, \overline{N}_i \gamma^\mu P_L \nu_\alpha\, Z_\mu + \text{h.c.}}$$

LNH

The bare term in the file is

$$\texttt{LNHbare} = -\sum_{i=1}^{3} \sum_{\alpha=e,\mu,\tau} \frac{g_N m_{N_i}}{2M_W} V_{\alpha i}\, \overline{N}_i P_L \nu_\alpha\, H .$$

The Lagrangian term is

$$\boxed{\texttt{LNH} = -\sum_{i=1}^{3} \sum_{\alpha=e,\mu,\tau} \frac{g_N m_{N_i}}{2M_W} V_{\alpha i}\, \overline{N}_i P_L \nu_\alpha\, H + \text{h.c.}}$$

LNG

The neutral-Goldstone part of the bare term is

$$i \sum_{i=1}^{3} \sum_{\alpha=e,\mu,\tau} \frac{g_N m_{N_i}}{2M_W} V_{\alpha i}\, \overline{\nu_\alpha} P_R N_i\, G^0 .$$

The charged-Goldstone part of the bare term is

$$i \sum_{i=1}^{3} \sum_{\alpha=e,\mu,\tau} \frac{g_N m_{N_i}}{\sqrt{2}M_W} V_{\alpha i}\, \overline{\ell_\alpha} P_R N_i\, G^- .$$

Thus

$$\boxed{\texttt{LNG} = i \sum_{i=1}^{3} \sum_{\alpha=e,\mu,\tau} \frac{g_N m_{N_i}}{2M_W} V_{\alpha i}\, \overline{\nu_\alpha} P_R N_i\, G^0 + i \sum_{i=1}^{3} \sum_{\alpha=e,\mu,\tau} \frac{g_N m_{N_i}}{\sqrt{2}M_W} V_{\alpha i}\, \overline{\ell_\alpha} P_R N_i\, G^- + \text{h.c.}}$$

Field Table

`.fr` class	Symbol	Spin	SU(3) rep	SU(2) rep	$U(1)$ charge / hyper-charge	Self-conjugate	Mass
`F[131]`	`N1`	1/2	singlet, no color index declared	singlet, no weak index declared	$Q = 0,\ Y = 0$	yes	`mN1 = 300.`
`F[132]`	`N2`	1/2	singlet, no color index declared	singlet, no weak index declared	$Q = 0,\ Y = 0$	yes	`mN2 = 500.`
`F[133]`	`N3`	1/2	singlet, no color index declared	singlet, no weak index declared	$Q = 0,\ Y = 0$	yes	`mN3 = 1000.`

Widths are also declared:

$$\texttt{WN1} = 0.303, \qquad \texttt{WN2} = 1.50, \qquad \texttt{WN3} = 12.3.$$

Parameters

Parameter	Type	Value	Multiplies	Physical meaning
`VeN1`	external, real	`1.0`	N_1-electron-family terms in `LNCC`, `LNNC`, `LNH`, `LNG`	mixing coefficient between N_1 and the electron lepton family
`VeN2`	external, real	`0.0`	N_2-electron-family terms	mixing coefficient between N_2 and the electron lepton family
`VeN3`	external, real	`0.0`	N_3-electron-family terms	mixing coefficient between N_3 and the electron lepton family
`VmuN1`	external, real	`0.0`	N_1-muon-family terms	mixing coefficient between N_1 and the muon lepton family
`VmuN2`	external, real	`1.0`	N_2-muon-family terms	mixing coefficient between N_2 and the muon lepton family
`VmuN3`	external, real	`0.0`	N_3-muon-family terms	mixing coefficient between N_3 and the muon lepton family
`VtaN1`	external, real	`0.0`	N_1-tau-family terms	mixing coefficient between N_1 and the tau lepton family
`VtaN2`	external, real	`0.0`	N_2-tau-family terms	mixing coefficient between N_2 and the tau lepton family

Parameter	Type	Value	Multiplies	Physical meaning
`VtaN3`	external, real	`1.0`	N_3-tau-family terms	mixing coefficient between N_3 and the tau lepton family

The file also defines one internal parameter:

$$\texttt{gN} = \frac{\texttt{ee}}{\texttt{sw}},$$

so the new interactions use the Standard Model weak $SU(2)_L$ coupling normalization.

Physics Summary

The model adds three electrically neutral, hypercharge-zero, self-conjugate spin-$\frac{1}{2}$ fermions N_1, N_2, N_3 to the Standard Model. They are gauge singlets at the level of their kinetic terms, but interact through mixing-suppressed charged-current, neutral-current, Higgs, and Goldstone couplings to the SM lepton families.

It mediates processes such as production and decay of heavy neutral leptons through $W^{\pm}$, Z, and Higgs exchange, including channels $N_i \leftrightarrow \ell_\alpha W$, $N_i \leftrightarrow \nu_\alpha Z$, and $N_i \leftrightarrow \nu_\alpha H$, with rates controlled by $V_{\alpha i}$ and the masses m_{N_i}.

Paper cross-check

Comparison of `reconstruction.md` Against the Paper Model

Paper Locations

The paper defines the heavy-neutrino model in the **"HEAVY NEUTRINO MODEL"** section.

Relevant definitions:

- **Eq. (7):** chiral neutrino fields are rotated into light and heavy mass eigenstates using blocks $U_{3\times3}$, $V_{3\times n}$, $X_{n\times3}$, $Y_{n\times n}$.
- **Eq. (8):** the flavor neutrino state is

$$\nu_\ell = \sum_{m=1}^{3} U_{\ell m}\nu_m + \sum_{m'=1}^{n} V_{\ell m'} N^c_{m'}.$$

- Immediately after Eq. (8), the paper states: **"For simplicity, we consider only one heavy mass eigenstate, labeled by N."**
- **Eq. (9):** the electroweak interaction Lagrangian with W, Z, and h is given.
- **"COMPUTATIONAL SETUP" section:** the paper says the above Lagrangian is implemented **"with Goldstone boson couplings in the Feynman gauge"**, but does not print the Goldstone terms explicitly.

Term-by-Term Comparison

paper term (eq. ref)	reconstruction term	verdict	notes
Heavy-light mixing rotation $(\nu_L, N_R^c)^T = \begin{pmatrix} U & V \\ X & Y \end{pmatrix} (\nu_m, N_{m'}^c)^T$, Eq. (7)	Three independent self-conjugate neutral fermions N_i, $i = 1, 2, 3$, with mixing parameters $V_{\alpha i}$	disagree	The paper gives the general n-heavy-state rotation but then specializes the analysis to **one** heavy mass eigenstate N. The reconstruction keeps three heavy states N_1, N_2, N_3.
Flavor state $\nu_\ell = \sum_m U_{\ell m} \nu_m + \sum_{m'} V_{\ell m'} N_{m'}^c$, Eq. (8)	Uses ℓ_α, ν_α, and $V_{\alpha i}$, but does not reconstruct the $U_{\ell m}$ light-neutrino mixing part	missing-in-reconstruction	The reconstruction captures active-heavy mixing $V_{\alpha i}$, but omits the explicit light-neutrino PMNS block $U_{\ell m}$ appearing in the paper's field definition.
Paper specialization to one heavy mass eigenstate N, text after Eq. (8)	Field table contains N_1, N_2, N_3 with masses 300, 500, 1000 GeV and widths 0.303, 1.50, 12.3	disagree	The reconstruction appears to describe a three-heavy-neutrino implementation or benchmark file, not the one-heavy-state simplified model printed in the paper.
New heavy field is a neutral heavy neutrino mass eigenstate N, Eqs. (7)-(9)	N_i are neutral, color-singlet, weak-singlet, hypercharge-zero, self-conjugate spin-$\frac{1}{2}$ fields	agree	The singlet Majorana interpretation is consistent with the Type-I-like heavy-neutrino setup and the use of N^c in the paper, aside from the multiplicity mismatch.
Heavy-neutrino kinetic and mass terms, implicit in the mass-eigenstate setup around Eqs. (7)-(9)	$\sum_i \left[\frac{i}{2} \bar{N}_i \gamma^\mu \partial_\mu N_i - \frac{1}{2} m_{N_i} \bar{N}_i N_i \right]$	agree	The Majorana normalization is standard. The paper does not print this term explicitly, so agreement is with the implied field content, not with a displayed equation.
Light-neutrino charged-current term $-\frac{g}{\sqrt{2}} W_\mu^+ \sum_{\ell=e}^{\tau} \sum_{m=1}^{3} \bar{\nu}_m U_{\ell m}^* \gamma^\mu P_L \ell +$ H.c., Eq. (9)	Not included except through a generic $\mathcal{L}_{\rm SM}$ placeholder	missing-in-reconstruction	If $\mathcal{L}_{\rm SM}$ is meant to include the PMNS-rotated light-neutrino charged current, this may be only a documentation omission. As written, the reconstruction does not state the $U_{\ell m}$ structure.

paper term (eq. ref)	reconstruction term	verdict	notes
Heavy-neutrino charged-current term $-\frac{g}{\sqrt{2}} W_\mu^+ \sum_{\ell=e}^{\tau} \bar{N}^c V_{\ell N}^* \gamma^\mu P_L \ell^- +$ H.c., Eq. (9)	$+\frac{g_N}{\sqrt{2}} \sum_{i,\alpha} V_{\alpha i} \bar{N}_i \gamma^\mu P_L \ell_\alpha W_\mu +$ h.c., with $g_N = e/s_w$	disagree	Coupling magnitude and chirality agree if $g_N = g$, V is real, and $N = N^c$. Differences: sign, V^* vs real V, W^+ charge labeling suppressed, and three N_i instead of one N.
Light-neutrino neutral-current term $-\frac{g}{2c_W} Z_\mu \sum_{\ell=e}^{\tau} \sum_{m=1}^{3} \bar{\nu}_m U_{\ell m}^* \gamma^\mu P_L \nu_\ell +$ H.c., Eq. (9)	Not included except through generic $\mathcal{L}_{\mathrm{SM}}$ placeholder	missing-in-reconstruction	The reconstruction does not spell out the PMNS light-neutrino neutral current present in the paper's Eq. (9).
Heavy-neutrino neutral-current term $-\frac{g}{2c_W} Z_\mu \sum_{\ell=e}^{\tau} \bar{N}^c V_{\ell N}^* \gamma^\mu P_L \nu_\ell +$ H.c., Eq. (9)	$+\frac{g_N}{2c_w} \sum_{i,\alpha} V_{\alpha i} \bar{N}_i \gamma^\mu P_L \nu_\alpha Z_\mu +$ h.c.	disagree	Coupling magnitude and left chirality agree under $g_N = g$, real V, and Majorana N. Differences are the overall sign, missing complex conjugation, and three heavy states rather than the paper's one-state simplification.
Heavy-neutrino Higgs term $-\frac{g m_N}{2M_W} h \sum_{\ell=e}^{\tau} \bar{N}^c V_{\ell N}^* P_L \nu_\ell +$ H.c., Eq. (9)	$-\sum_{i,\alpha} \frac{g_N m_{N_i}}{2M_W} V_{\alpha i} \bar{N}_i P_L \nu_\alpha H +$ h.c.	disagree	Coefficient, mass proportionality, chirality, and sign match for one real-mixing Majorana state. Differences are V^* vs real V, N^c vs N_i, and three heavy states instead of one.
Goldstone boson couplings included in Feynman gauge, "COMPUTATIONAL SETUP" section	Neutral Goldstone $i\frac{g_N m_{N_i}}{2M_W} V_{\alpha i} \bar{\nu}_\alpha P_R N_i G^0 +$ h.c.; charged Goldstone $i\frac{g_N m_{N_i}}{\sqrt{2}M_W} V_{\alpha i} \bar{\ell}_\alpha P_R N_i G^- +$ h.c.	agree	The paper states Goldstone couplings are implemented but does not print them. The reconstruction has the standard Feynman-gauge mass-proportional Goldstone completion of the heavy-neutrino interactions, modulo the same one-vs-three and real-V issues.

paper term (eq. ref)	reconstruction term	verdict	notes
Cross-section factorization $\sigma(pp \to NX) = \lvert V_{N\ell}\rvert^2 \sigma_0(pp \to NX)$, Eq. (10)	Mixing parameters $V_{\alpha i}$ multiply all $W/Z/h/G$ interactions; benchmark values are diagonal $V_{eN1} = V_{\mu N2} = V_{\tau N3} = 1$	agree	The reconstruction's interactions are linear in $V_{\alpha i}$, so rates factorize as $\lvert V_{\alpha i}\rvert^2$ for single-coupling channels. The benchmark choice with three unit mixings is not the one-heavy-state simplification used in the paper text.
Paper model uses SM weak coupling g, Eq. (9)	Defines $g_N = e/s_w$	agree	This is the usual electroweak identity $g = e/\sin\theta_W$.

Disagreements and Checks

issue	severity	what a human should check
The paper specializes to one heavy mass eigenstate N, while the reconstruction contains three N_i.	substantive	Check whether `sanitized.fr` intentionally implements a three-state generalization of the public model file, or whether the paper comparison should be restricted to a single selected state such as `n2`.
The reconstruction omits the explicit light-neutrino $U_{\ell m}$ charged- and neutral-current terms from Eq. (9).	substantive	Check whether these terms are supplied by the imported SM/FeynRules base model or whether the implementation drops PMNS-rotated light-neutrino interactions.
The charged-current and neutral-current heavy-neutrino terms have the opposite displayed overall sign from Eq. (9).	convention	Check the FeynRules sign convention, field ordering, and whether a field or mixing-parameter phase redefinition makes the signs equivalent in generated vertices.
The reconstruction treats $V_{\alpha i}$ as real, while the paper writes $V^*_{\ell N}$.	substantive	Check whether the implementation is restricted to real active-heavy mixing or whether complex phases are supported elsewhere but lost in the reconstruction.
The reconstruction uses N_i directly, while Eq. (9) writes N^c.	convention	For Majorana heavy neutrinos this is usually equivalent, but check the implementation's Majorana/self-conjugate declaration and fermion-flow conventions.

issue	severity	what a human should check
The reconstruction gives explicit masses and widths for three heavy states, while the paper discusses results as functions of a single m_N.	substantive	Check whether the benchmark masses and widths are implementation defaults rather than part of the paper's analytical model definition.
The Goldstone terms are reconstructed explicitly, but the paper only states that Goldstone couplings are included in Feynman gauge.	cosmetic	Check the actual model file or generated Feynman rules if an exact sign-level validation of Goldstone vertices is required.

Overall Assessment

The reconstruction captures the core heavy-neutrino interaction structure of the paper: neutral Majorana singlet fermions coupled to SM leptons through active-heavy mixing, with the expected W, Z, Higgs, and Feynman-gauge Goldstone interactions and the correct weak-coupling normalization. The largest mismatch is scope: the paper's displayed model specializes to one heavy mass eigenstate N, while the reconstruction describes three benchmark heavy states with a matrix of real mixings. The reconstruction also suppresses or delegates the light-neutrino $U_{\ell m}$ sector from Eq. (9), and it loses the paper's complex conjugation on $V_{\ell N}$. Most sign and N versus N^c differences may be convention-dependent for Majorana fields, but the heavy-state multiplicity and real-versus-complex mixing assumptions are physics-level differences a reviewer should verify against the actual implementation intent.

Suggested checks for the reviewer

1. Every verbatim `.fr` term above has a reconstructed LaTeX counterpart with the same field content, chirality and conjugation.
2. Kinetic terms: covariant derivative gauge content matches the field's representations; normalization is canonical.
3. Non-self-conjugate interaction terms appear together with their Hermitian conjugates.
4. Quantum numbers in the field table match the `.fr` declarations (and the paper, where the cross-check table flags disagreements).
5. Numeric masses/couplings are placeholders unless the paper pins them — treat values as demo inputs, not measurements.
6. Sanitizer scope: 13 prose labels were scrubbed; field/parameter symbol names were kept and may hint at the model's identity.

Physicist sign-off

- Reviewed by: ________________________________ Date: __________________
- Verdict: [ ] approve [ ] approve with corrections [ ] reject
- Notes: